Hao Chen

Gender M | Date of birth 18/12/1994 | Nationality China
hao.chen@outlook.it
0039 338 3677077

Research Profile

Experimental researcher in functional oxides, uncovering structure–property relationships through synchrotron X-ray spectroscopy and advanced characterization techniques, aimed at enabling the rational design of functional devices.

Employment & Education

Postdoctoral Fellow – Department of Physics, Politecnico di Milano, Italy	05/2024–11/2025
Ph.D. in Applied Physics – Department of Physics, Politecnico di Milano, Italy	11/2020–05/2024
Research stay at European Synchrotron Radiation Facility (ESRF), France	09/2022–12/2022
M.Sc. in Materials Engineering and Nanotechnology, Politecnico di Milano, Italy	09/2017–07/2020
B.Sc. in Materials Science and Engineering, Henan Polytechnic University, China	09/2013–06/2017

Professional Experience

WO_{3-x} Thin Films

- Quantified defect concentrations using EDXS, XPS, and Auger spectroscopy via collaborative efforts.
- Developed a graded ellipsometry model to resolve depth-dependent dielectric profiles beyond uniform approximations.
- Applied XANES/EXAFS to correlate electronic structure and oxidation states with local coordination and oxygen vacancies.
- Implemented Python-based NNMF analysis to decompose large RIXS datasets and identify hidden spectral components.
- Established SEM/AFM protocols to capture nanoscale morphology and surface roughness.
- Correlated XRD and Raman spectroscopy to resolve amorphous-to-crystalline structural transitions.
- Quantified electrical transport (van der Pauw), linking conductivity to oxygen vacancy defects.
- Probed ultrafast carrier dynamics using transient absorption (pump–probe) spectroscopy in collaboration with spectroscopy specialists.

SrO/$SrIrO_3$ Superlattices & High-Entropy Oxide Thin Films

- Optimized PLD growth for epitaxy and interface control in complex oxides within a collaborative research framework.
- Resolved film periodicity and interface sharpness via combined XRD/XRR analysis.
- Validated composition using complementary XPS and SEM–EDXS.

Novel Cathode Material for Solid Oxide Fuel Cell

- Performed operando XAS to track real-time electronic & structural evolution under working conditions.
- Developed Python workflows to decouple thermal and chemical effects in operando measurements.

Research Proposals

Co-authored proposals securing access to major user facilities: ESRF, Elettra Synchrotron, PoliFab.

Languages

English: C1 – spoken and written proficiency. **Italian**: A2 – basic communication. **Chinese**: native.

Supervision

Co-supervised a M.Sc. student.